

A1: Needfinding Review

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Team Members



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BioE ‘26
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01

Domain Selection









Our studio: Design for Meaningful work

So we asked ourselves...







What makes work feel meaningful?

Output alone isn't enough.

It's the people and the why.







What threatens that feeling today?

AI → output without effort

Leaves users to prompt, monitor, verify







Where would that loss hurt the most?

And what does that mean for
“meaningful work”?






Creativity
...
Who within creativity?

Craft ↔ Livelihood Spectrum

People learning high-barrier tools



Lifelong practitioners



Craft on the side

Craft is the livelihood

Shared by both ends: our hunches, not findings

Struggle Routine Feedback Pull toward AI

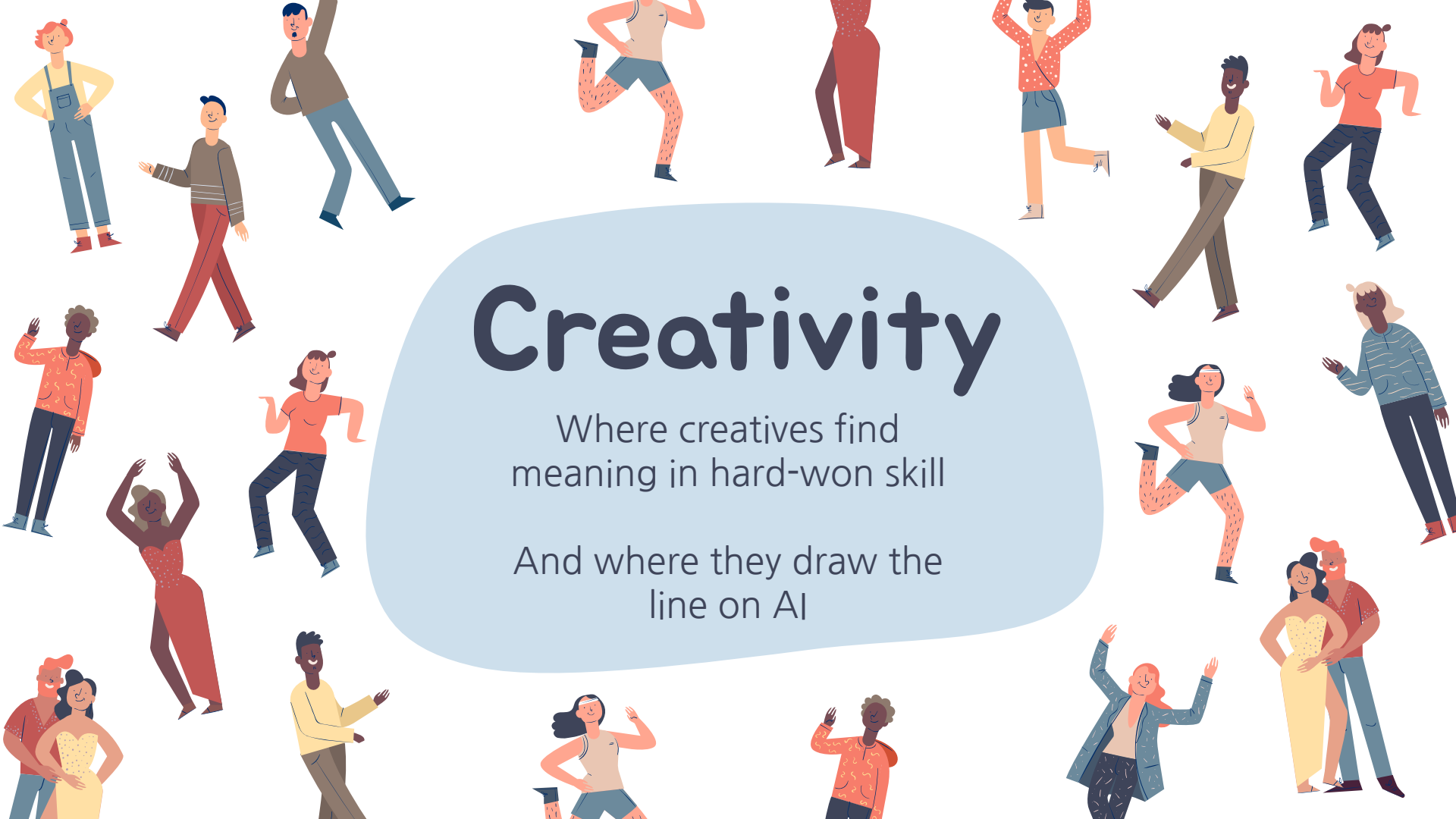


Why them?

Skills take years to build.
AI offers a way around building.






Creativity
Where creatives find
meaning in hard-won skill
And where they draw the
line on AI

02

Needfinding Methodology









Recruitment Strategy



Personal Network:

Former instructor

Composer of piece performed

TA

Targeted Referrals

Requested friend to find someone with Photoshop experience

Requested fellow interns

Selection Priorities: Range of Experiences

Medium

UX, graphic design, illustration, poetry, etc.

1



Skill Stage

Hobbyist, student, early-career, industry, professionals, etc.

2

Craft vs. Livelihood

Side pursuit to a whole career

3



Selection Priorities: Diversity

Setting

Stanford, UCB, industry

5

AI exposure

Heavy pressure vs. none

Field

STEM vs. arts and humanities

4

6





Five Interviewees



Alex

SWE
Painter & drawer
Stanford Alumni



Kevin

Stanford English lecturer
Former technical writer
Poet, translator



Gary

Stanford composition professor



Christina

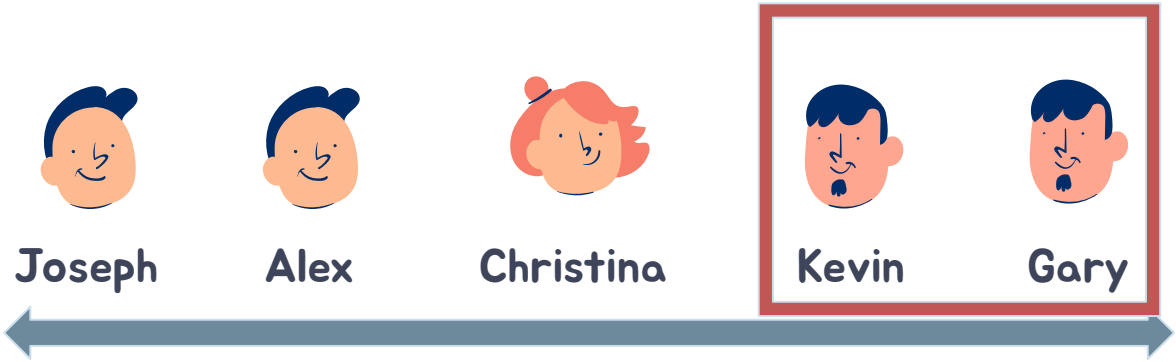
UI/UX Intern
Graphic design
Grad Student



Joseph

Integrative biology @ UCB
Photoshop, Clip Studio, Maya

Craft ↔ Livelihood Spectrum



G: craft is his whole life, 30+ years

K: deliberately left a tech career for poetry

Extreme in what way?

How completely creative craft has become their livelihood, and for how long

Where?











Interview Logistics: Apparatus



1

Audio Recording
Voice memos

2

Transcribe
Online softwares available

3

Zoom
For long-distance interview



Roles in Each Interview

We each led interviews at least once

With

- Brian taking the lead for Christina, Kevin, and Gary
- Cherie for Alex
- Ryan for Joseph

While the remaining members recorded and took notes







Interview Script and Questions

Understanding where meaning lives before asking about tools

Context

Who they are and where creativity lives

Learning Journey

How the skill was built, and where it almost broke

Process

Day-to-day work and getting unstuck

AI

Where it shows up, and where they draw the line

Meaningful work

The same closing q for all five

Each guide was tailored to the person. Every interview closed with the same question



Tailored to each interviewee

Each guide included questions written around the person's own craft

Joseph - illustrator

"If you could improve the onboarding of a tool like Photoshop, what would you change?"

Alex - engineer & painter

"Has your passion for painting shaped why you chose the graphics side of software?"

Christina - UX designer

"Looking back, what would you have wanted to know as a student before entering industry?"

Kevin - poet and translator

"Your work includes translating poems. Where do originality and creativity come in?"

Gary - composer

"Walk me through how the Grateful Dead orchestra project started"

Context and Background

Who they are, and where creative work lives in their lives



"Could you give a brief introduction of yourself, your background, and your experience with art?"

Alex - engineer & painter

"Tell us about your poetry. What do you write, and where does that live in your life?"

Kevin - poet and translator

"Where do you find yourself engaging in art the most: classes, extracurriculars, or personal time?"

Joseph - illustrator

Learning journey And obstacles

How the skill was built, and where it almost broke



"Tell me about a time a composition wasn't coming together. What did you try?"

Gary - composer

"Can you walk us through your process of learning Photoshop?"

Joseph - illustrator

"Have you experienced any struggles accessing creative learning, for painting or for graphics?"

Alex - engineer & painter

Process and getting unstuck

What the work looks like day to day, and what happens when it stalls



"How do you know when a piece is finished?"

Gary - composer

"When you hit roadblocks, in ideas or energy, how do you navigate them?"

Kevin - poet and translator

"Under what circumstances do you feel most creative? What inspires you?"

Joseph - illustrator

AI in their craft

Where AI already shows up, and where they draw the line



"How has AI impacted your work, if at all?"

Joseph and Alex

"Do you feel pressure at work around AI? Do your values align with your company's?"

Christina - UX designer

"Is there any part of your craft you'd never want AI to be involved in? Why?"

Christina, Joseph, Kevin



And...

"When you hear the term 'meaningful work,' what does that mean to you?"

03

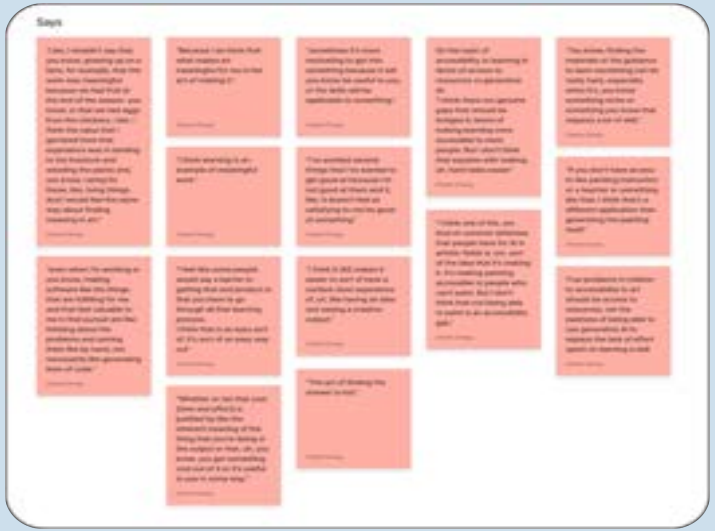
Interview Results & Analysis







Empathy Map - Alex



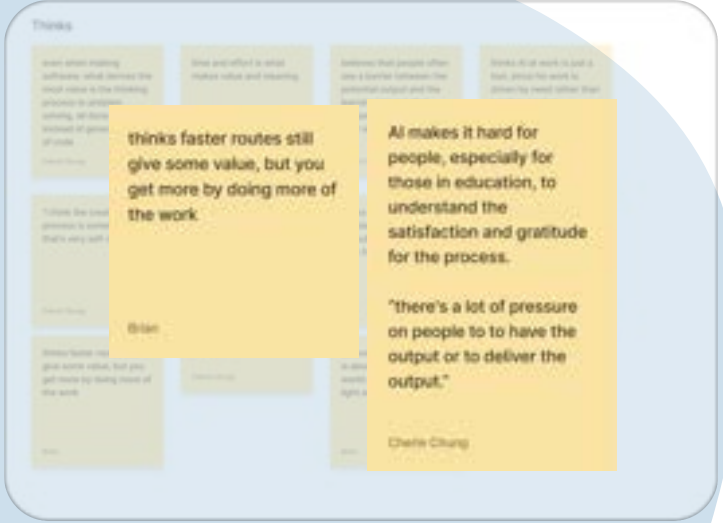


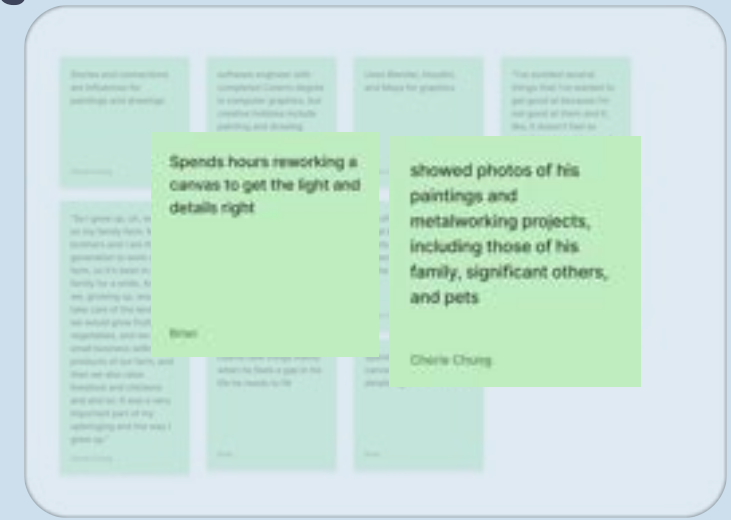


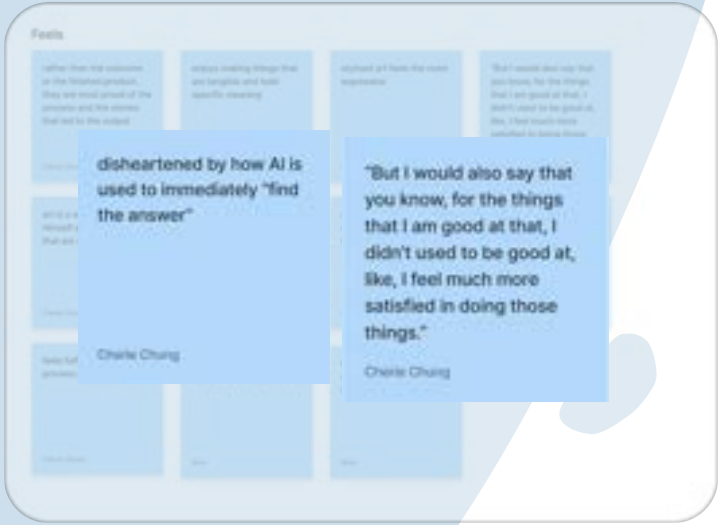


Key Takeaways - Alex









Analysis - Alex

Quote: “But I would also say that you know, for the things that I am good at that, I didn't used to be good at, I feel much more satisfied in doing those things.”

Insight: Measures meaning and fulfillment from the process, more so than the output

- The learning, the struggling, and the problem-solving is where the value of the final output comes from
- If you bypass that process, you have an output, which is valuable, but that’s it

Need: Reason to struggle rather than have immediate output/instant gratification; a reason to sacrifice valuable time

Empathy Map - Christina

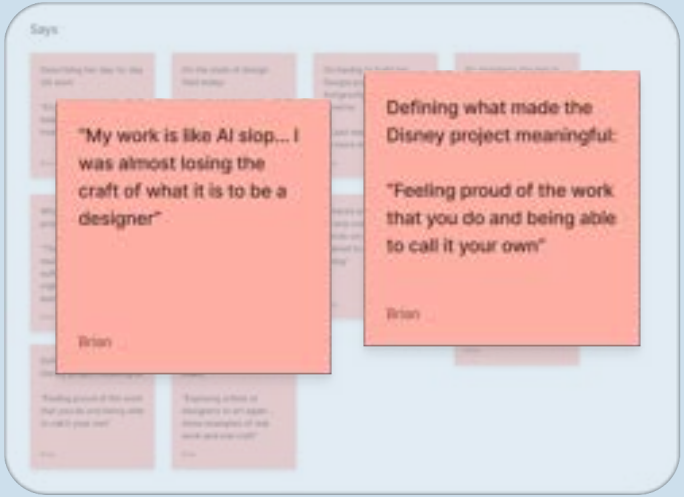


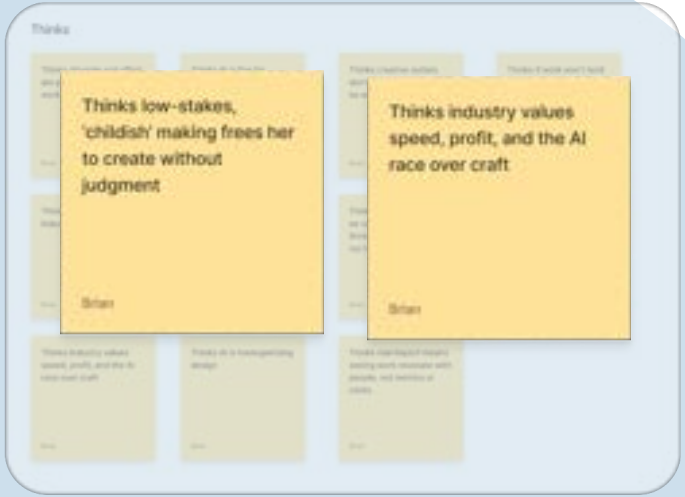


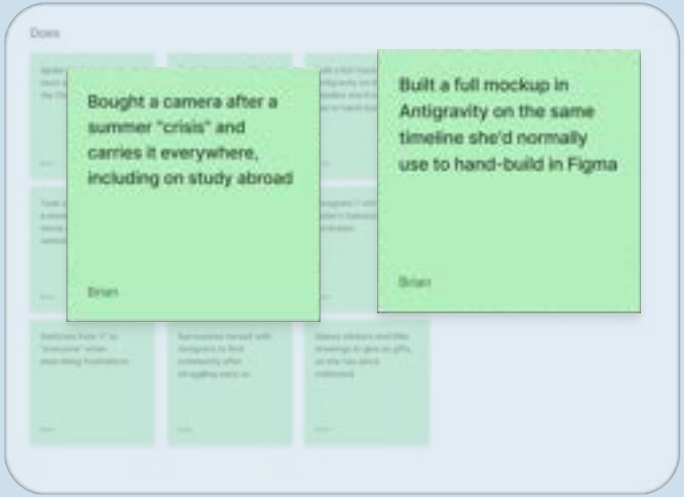


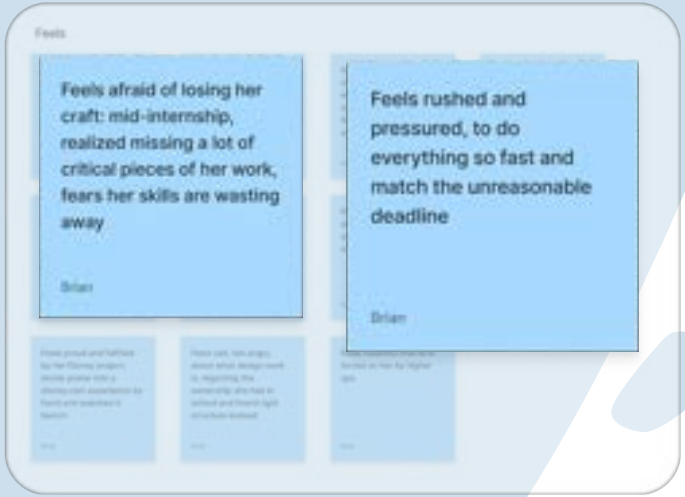


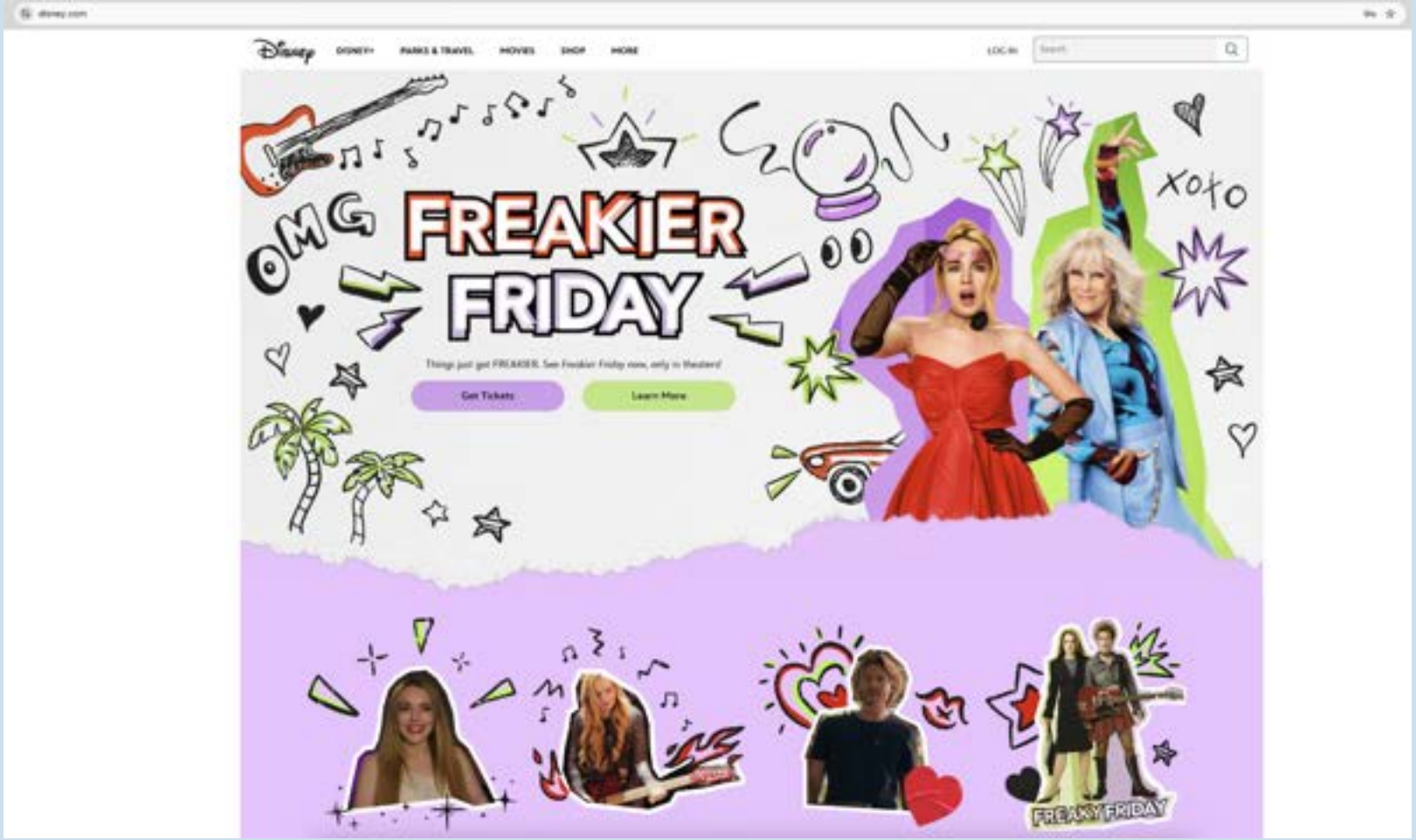
Key Takeaways - Christina










Disney
DISNEY+
PARKS & TRAVEL
MOVIES
SHOP
MORE
LOG IN
OMG
FREAKIER
FRIDAY
Things just got FREAKIER. See Freakier Friday now, only in theaters!
Get Tickets
Learn More
xoxo
FREAKY FRIDAY

Analysis - Christina

Surprise: AI didn't buy her any time for craft. @Google, asked to produce a polished, end-to-end mockup in Antigravity on the same timeline normally used to build by hand in Figma → AI SLOP

Insight: Measures meaning by **ownership**: being able to "call it your own"

- When company turned AI's speed into **higher output expectations,** removed the hands-on steps where ownership lived → felt craft slipping
- After summer 'crisis', rebuilt it outside work through small, low-stakes making: a camera she carries everywhere & stickers as gifts

Need: a way to keep a small, hands-on creative practice alive alongside demanding work, without it becoming another metric or deadline

Empathy Map - Kevin

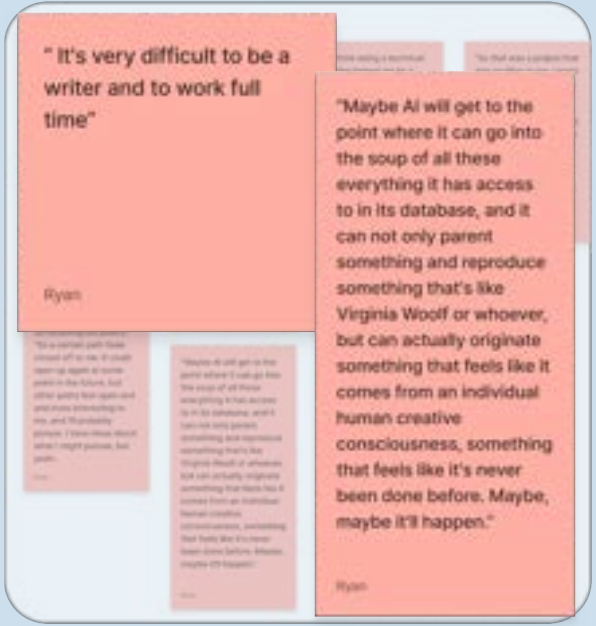


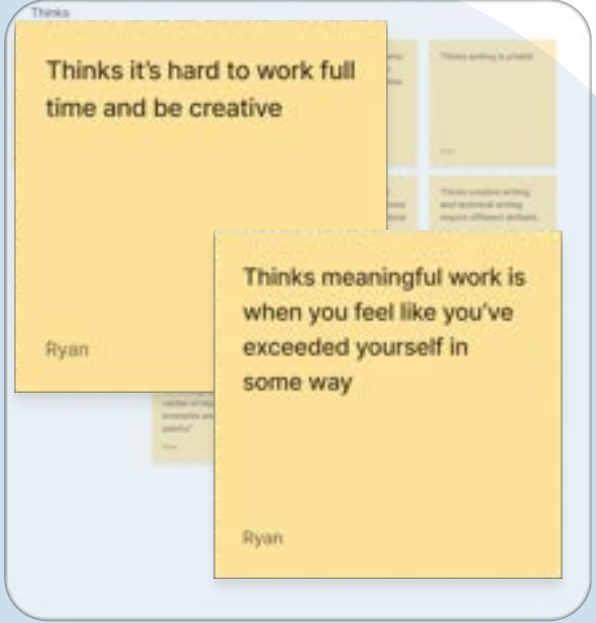




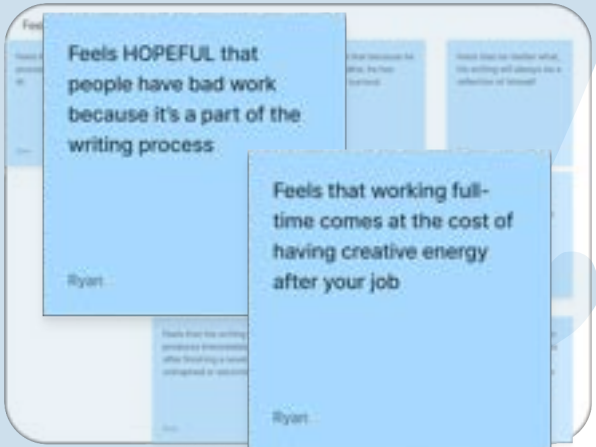


Key Takeaways - Kevin









Up and Down Picacho Peak

Hunter trail, I walk the path
named for the Confederate Captain
who burned haystacks to starve Union horses.
Blue lupine guards the way, globe mallows

beneath my feet map the landscape's
vastness: both sides were lost,
hadn't meant to skirmish. If they almost died
of thirst, there were too many reminders

of water: palo verde swim in yellow and bees,
the peak curls like a wave about to break.
An easy climb, straight up through granite cliffs,
unlike this history. Slaves the territory

sought as soldiers to flush out Apache,
and keep the mines humming. I reach
the summit quickly, scenery and scent:
ploughed cotton fields, dead volcanoes.

Hiking Picacho Peak

I walk the path, Hunter trail,
named for the Confederate Captain
who burned haystacks to starve Union horses.
Blue lupine guards the way, globe mallows

mapped at my feet chart the landscape's
vastness: both sides were lost,
hadn't meant to skirmish. If they almost died
of thirst, there were too many reminders

of water: palo verde swim in yellow and bees,
the peak breaks like a wave, trail leading
straight up granite cliffs. I work my fingers
against stone, reach the summit quickly,

hands swollen. The territory wanted slaves,
soldiers to search out raiding Apaches.
Scenery and scent: cotton fields plowed out
beneath extinct volcanoes, I turn my back

Hiking Picacho Peak

I walk the only path, Hunter trail,
named for the Confederate Captain
who burned haystacks to starve Union horses.
Blue lupine guards the way, globe mallows
mapped at my feet chart the landscape's
vastness: both sides were lost here,
hadn't meant to skirmish. If they almost died
of thirst, there was much to remind them
of water: palo verde swim in yellow and bees,
the peak breaks like a wave, trail leading

straight up granite cliffs. I work my fingers
against stone, reach the peak quickly,
hands swollen. The territory wanted slaves,
soldiers to search out raiding Apaches.
Scenery and scent: cotton fields plowed out
beneath extinct volcanoes, I turn my back
on the freeway: the better idea of the West,
waterless arroyos toward the Sawtooth range,
no roads find it—beauty like stillness,
though it never lacks thorn and edge.

Analysis - Kevin

Quote: “ It's very difficult to be a writer and to work full time”

Insight: Full time work left him **too drained to write**: “you’re just tired”. Instead of quitting, he negotiated **part-time**, working three days at Oracle and writing on his days off, and the two **balanced well.** His desk job even gave him the discipline to write

- Creative work can survive a demanding job when it has protected, predictable time, not necessarily more time overall

Need: a way to **protect predictable time for writing** that a demanding job can’t crowd out

Empathy Map - Joseph









Key Takeaways - Joseph





Analysis - Joseph

Quote: “I’ll just copy it one for one. Repetitively, until I feel comfortable enough in the process that I can change it”

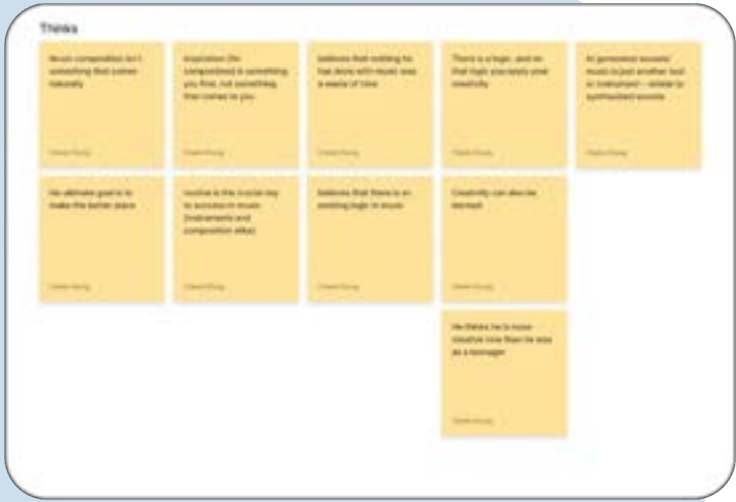
Insight: learns through **repetition before invention.** Copies a guide exactly until comfortable, fluency carries tool to tool.

- Value only shows **in hindsight:** Photoshop year felt ‘pointless’ at the time
- When drawing became an obligation in school, burnt out into art block

Need: a way to **build skills through small, guided repetition** and see their **progress** along the way, without art turning into an **obligation**

Empathy Map - Gary

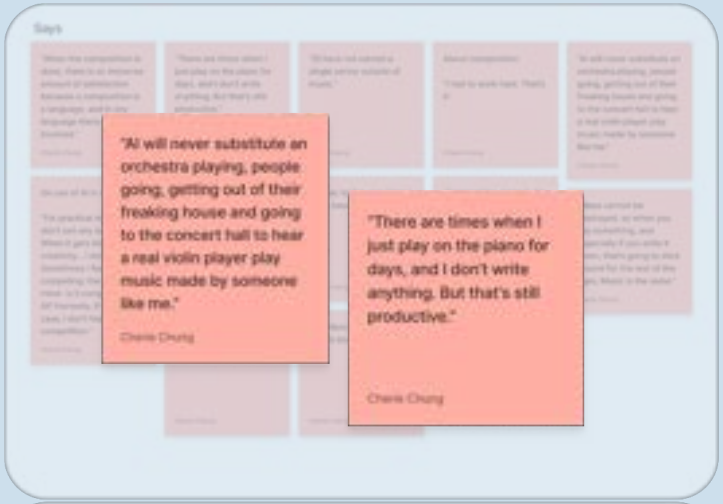


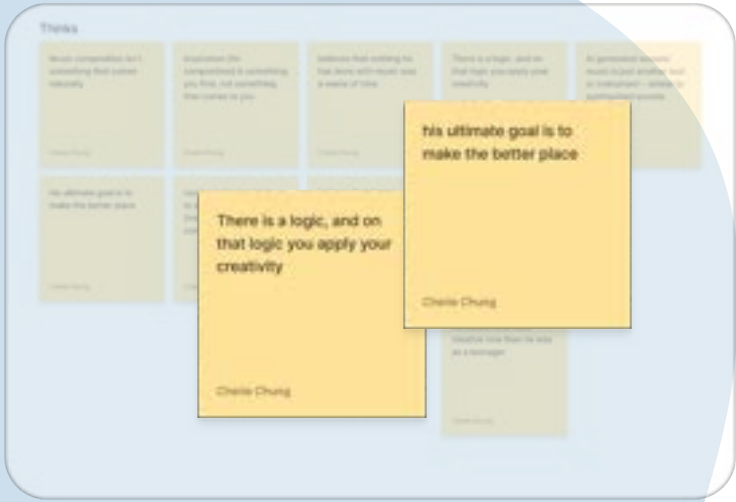


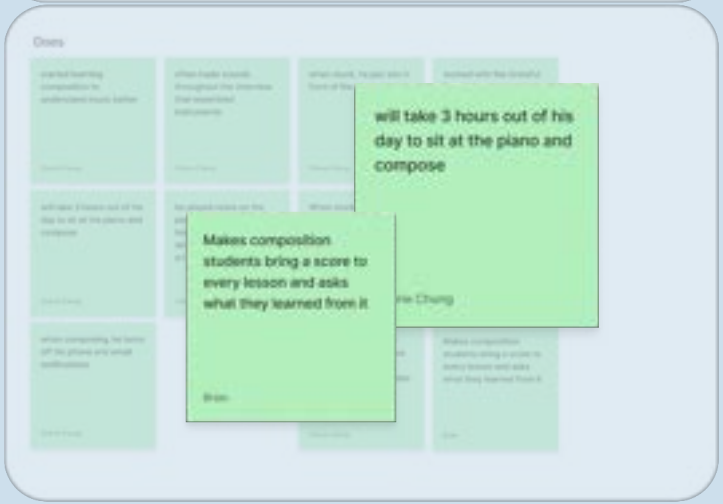


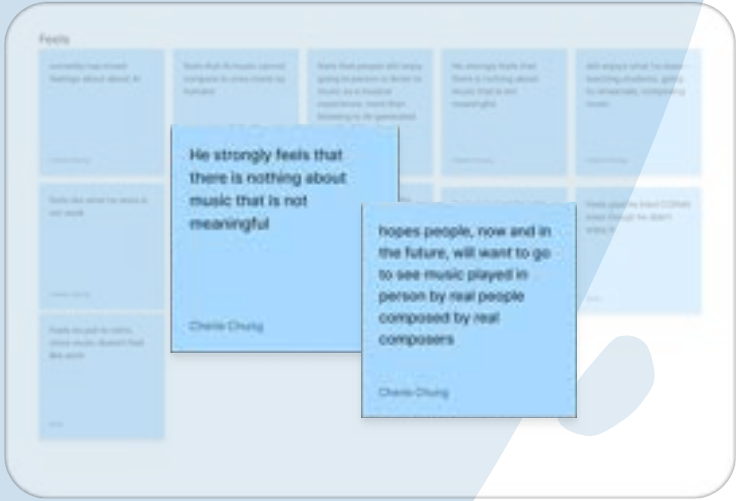


Key Takeaways - Gary









Analysis - Gary

Surprise: Gary was very neutral on the topic of AI and AI-generated music/composition despite stressing the importance of a composer's thorough understanding of music theory and obligation to set a composition routine

Insight: Music is a language he felt compelled to understand the underlying logic to then read and write, like you would any other language. Therefore, to get good at understanding music, he stressed the importance of a routine and practicing the "muscle of the language" [music].

Need: Routine and the **time and personal investment** to try and fail while composing

Reflection

Surprised by: We found that both of our “extreme” interviewees had positive and/or neutral views about AI usage in creative spaces, while the others had negative perspectives.

“For practical matters, I don’t see any issues. When it gets into creativity… I don’t know. Sometimes I feel like we’re competing; the human mind- is it competing with AI? Honestly, if that’s the case, I don’t fear that competition.”
- Gary

Reflection

Expected creatives would credit inspiration

… Instead, the experts credited routine

Kevin: credits the discipline he learned at his tech-writing desk
Gary: credits the everyday routine of just sitting down for hours and thinking

Reflection

Expected turning a passion into a job would deepen it

… Instead, it often drained it

Joseph: full-time art would be “depressing”

Christina: industry “tainted my passion”

Key Learnings (1)

Time is the real barrier, more than motivation or skill

Creative work gets crowded out by demanding work and school

- Kevin was too drained by full-time to write
- Christina asked why she has no time to pursue creativity during her work





Key Learnings (2)

Consistency beats intensity

Skill grows from small, regular practice, not bursts

- Gary: “it’s about routine,” students can’t compose by cramming
- Kevin wrote on fixed days each week
- Joseph builds fluency through repetition





Key Learnings (3)

Effort is the point

People want to make it themselves, not shortcut it

- Alex: AI makes it “easier to skip the meaningful part”
- Kevin: “you can’t have someone run the treadmill for you”
- Christina: called her work “AI slop”, goal of her design work was efficiency not creativity





Key Learnings (4)

Once it feels like an obligation, it stops working

- Joseph burned out on assigned drawing
- Christina lost meaning under deadlines and metrics
- Gary loves his work because it doesn't feel like work





What's Next?

Narrow the Problem

Focus on people trying to keep a creative practice alive alongside demanding work or school, where time is the real barrier.

Interview the Gaps

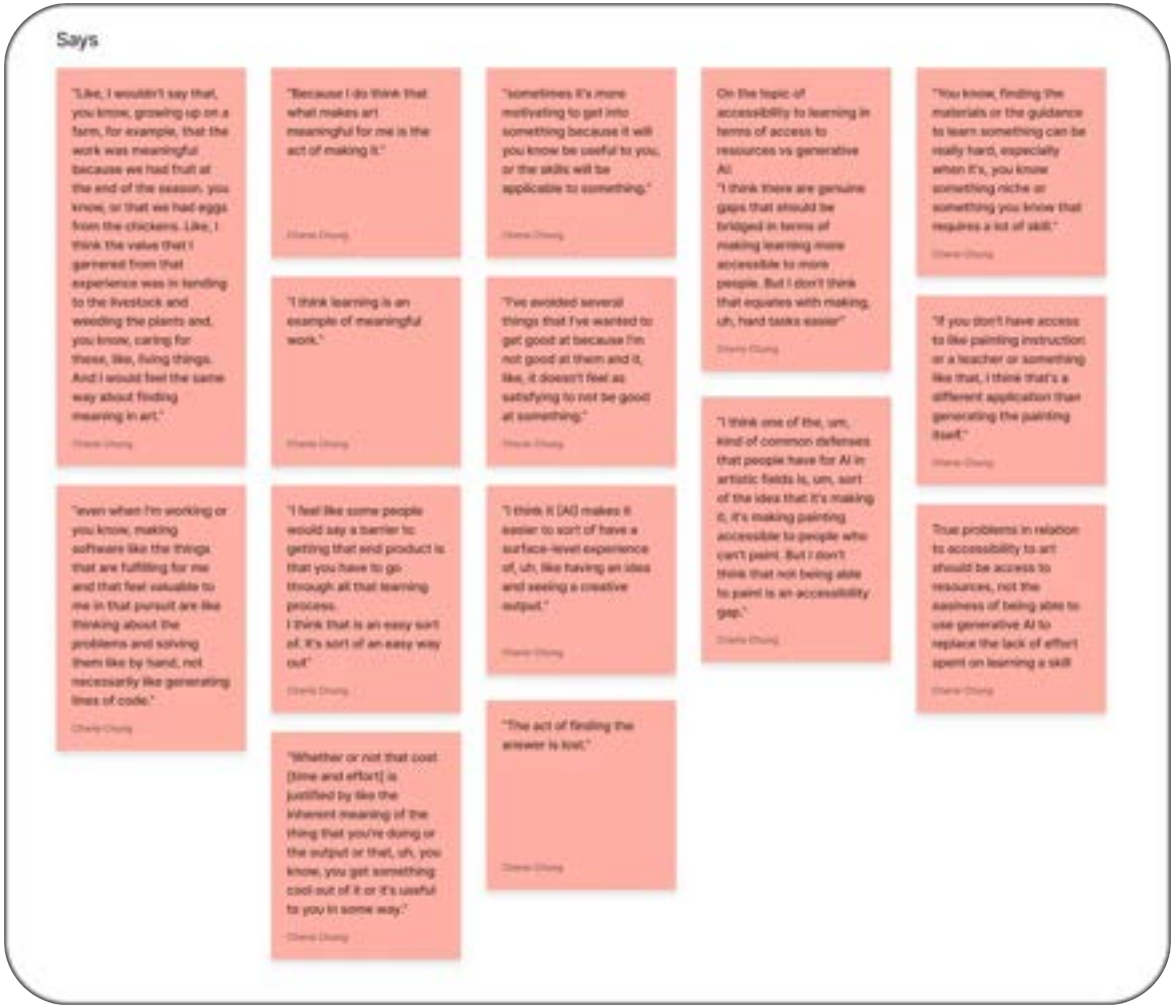
Talk to someone who started a creative habit and dropped it, and someone at the extreme: a daily maker or a heavy AI user.

A2

Turn our insights into POVs and HMWs, brainstorm solutions, and test whether short daily sessions, streaks, and prompts help or add pressure.


Appendix

Alex Empathy Map



Thinks

even when making software, what derives the most value is the thinking process to problem solving, all done by hand instead of generating lines of code

Cherie Chung

time and effort is what makes value and meaning

Cherie Chung

believes that people often see a barrier between the potential output and the learning process that happens in between the start and finish line.

Cherie Chung

thinks AI at work is just a tool, since his work is driven by need rather than creative passion

Brian

"I think the creative process is something that's very self-driven"

Cherie Chung

AI makes it hard for people, especially for those in education, to understand the satisfaction and gratitude for the process.

"there's a lot of pressure on people to to have the output or to deliver the output."

Cherie Chung

thinks people start hobbies chasing the output they see online, then hit the learning curve

Brian

considers usefulness is a good reason to start something, and meaning comes along the way

Brian

thinks faster routes still give some value, but you get more by doing more of the work

Brian

believes graphics, like art, is about capturing real-world phenomena like light and motion

Brian

Does

Stories and connections are influences for paintings and drawings

Charlie Chung

software engineer with completed Coterm degree in computer graphics, but creative hobbies include painting and drawing

Charlie Chung

Uses Blender, Houdini, and Maya for graphics

Charlie Chung

"I've avoided several things that I've wanted to get good at because I'm not good at them and it, like, it doesn't feel as satisfying to not be good at something."

Charlie Chung

"So I grew up, uh, working on my family farm. My brothers and I are the fifth generation to work on the farm, so it's been in my family for a while. And so we, growing up, would take care of the land and we would grow fruits and vegetables, and we ran a small business selling the products of our farm, and then we also raise livestock and chickens and and so. It was a very important part of my upbringing and the way I grew up."

Charlie Chung

showed photos of his paintings and metalworking projects, including those of his family, significant others, and pets

Charlie Chung

will often sit and ideate what kind of style he wants his work to follow depending on the subject of the painting/drawing

Charlie Chung

Studies reference paintings and photos in the style he wants before starting a piece

Brian

Learns new things mainly when he feels a gap in his life he needs to fill

Brian

Spends hours reworking a canvas to get the light and details right

Brian

Feels

rather than the outcome or the finished product, they are most proud of the process and the stories that led to the output

Cherie Chung

enjoys making things that are tangible and hold specific meaning

Cherie Chung

stylized art feels the most expressive

Cherie Chung

"But I would also say that you know, for the things that I am good at that, I didn't used to be good at, like, I feel much more satisfied in doing those things."

Cherie Chung

art is a way to express himself and capture things that are meaningful to him

Cherie Chung

frustrated by how AI is used to bypass the learning process

Cherie Chung

disheartened by how AI is used to immediately "find the answer"

Cherie Chung

Feels unsatisfied when he's bad at something new

Brian

feels fulfillment from the process of creation

Cherie Chung

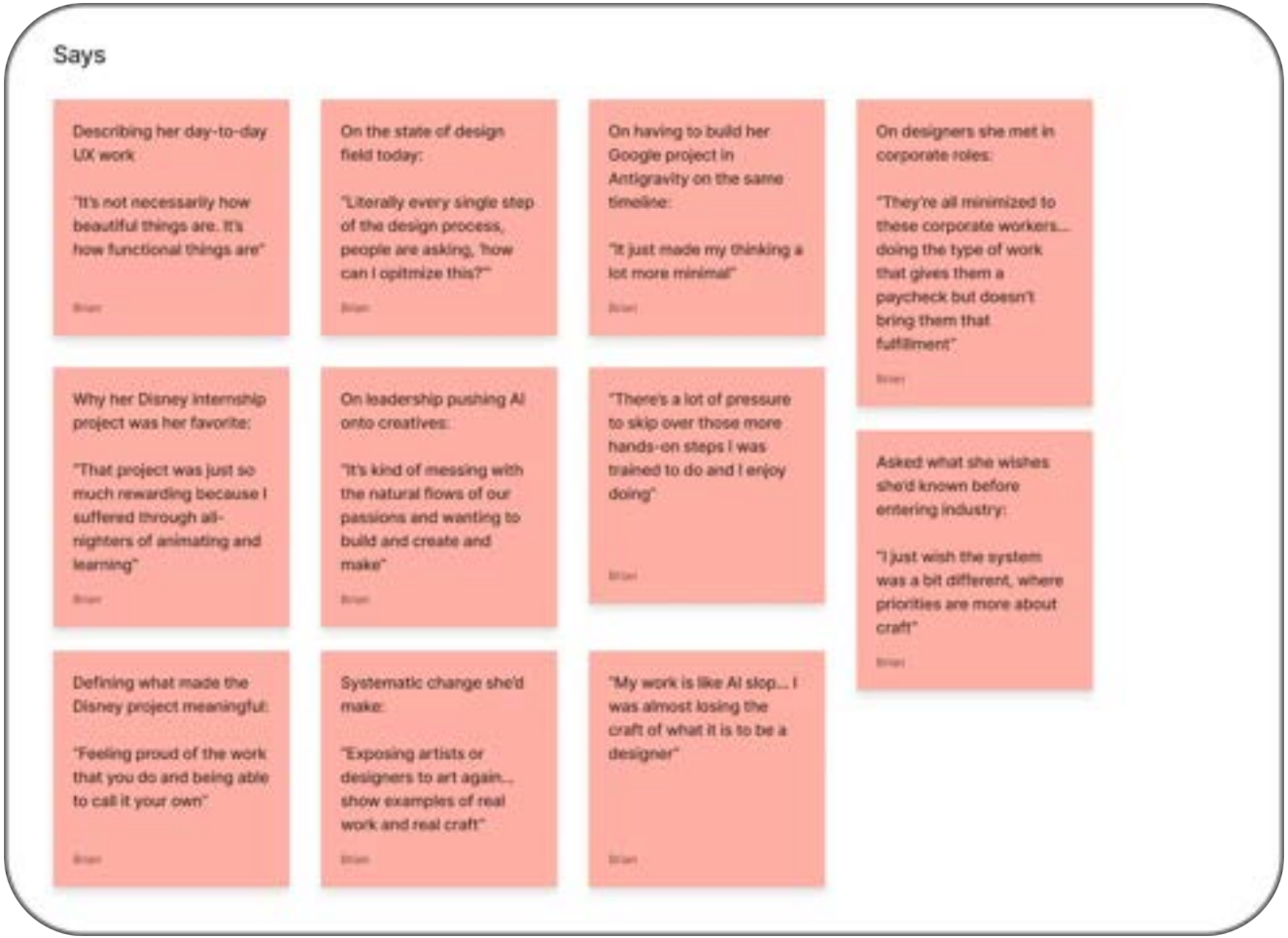
Feels short on time to pick up a new artistic practice

Brian

Feels fond of his farm upbringing and the lessons it taught him

Brian

Christina
Empathy Map



Thinks

Thinks struggle and effort are part of what makes work rewarding

Brian

Thinks AI is fine for implementation, but design decisions should stay human

Brian

Thinks creative outlets don't need a purpose to be worthwhile

Brian

Thinks if work won't hold her creativity, she has to find it somewhere else

Brian

Thinks creative agency in industry is an illusion

Brian

Thinks life shouldn't be all work

Brian

Thinks designers should be valued for how they think and solve problems, not how fast they ship

Brian

Thinks low-stakes, 'childish' making frees her to create without judgment

Brian

Thinks industry values speed, profit, and the AI race over craft

Brian

Thinks AI is homogenizing design

Brian

Thinks real impact means seeing work resonate with people, not metrics or clicks

Brian

Does

Spoke the longest and most animatedly about the Disney project

Brian

Bought a camera after a summer "crisis" and carries it everywhere, including on study abroad

Brian

Built a full mockup in Antigravity on the same timeline she'd normally use to hand-build in Figma

Brian

Spots AI-made portfolios, posters, and websites instantly and calls them out

Brian

Took over disney.com for a movie release, turning a movie poster into a website

Brian

Pulled all-nighters at Disney teaching herself animation

Brian

Designed T-shirts for her sister's Oakland nonprofit fundraiser

Brian

Switches from "I" to "everyone" when describing frustrations

Brian

Surrounded herself with designers to find community after struggling early on

Brian

Makes stickers and little drawings to give as gifts, as she has since childhood

Brian

Feels

Feels disillusioned with UX; year ago called herself very passionate about UX, and summer at Google changed that

Brian

Feels rushed and pressured, to do everything so fast and match the unreasonable deadline

Brian

Feels afraid of losing her craft: mid-internship, realized missing a lot of critical pieces of her work, fears her skills are wasting away

Brian

Feels unsure where she stands in the design world: design is broadly defined, that defining herself feels impossible

Brian

Feels proud and fulfilled by her Disney project; movie poster into a disney.com experience by hand and watched it launch

Brian

Feels BURNT OUT: feels like everything she does is work and work, work had crowded out everything she cared about

Brian

Feels moved by seeing direct impact: making t-shirts for her sister's nonprofit

Brian

Feels proud and fulfilled by her Disney project; movie poster into a disney.com experience by hand and watched it launch

Brian

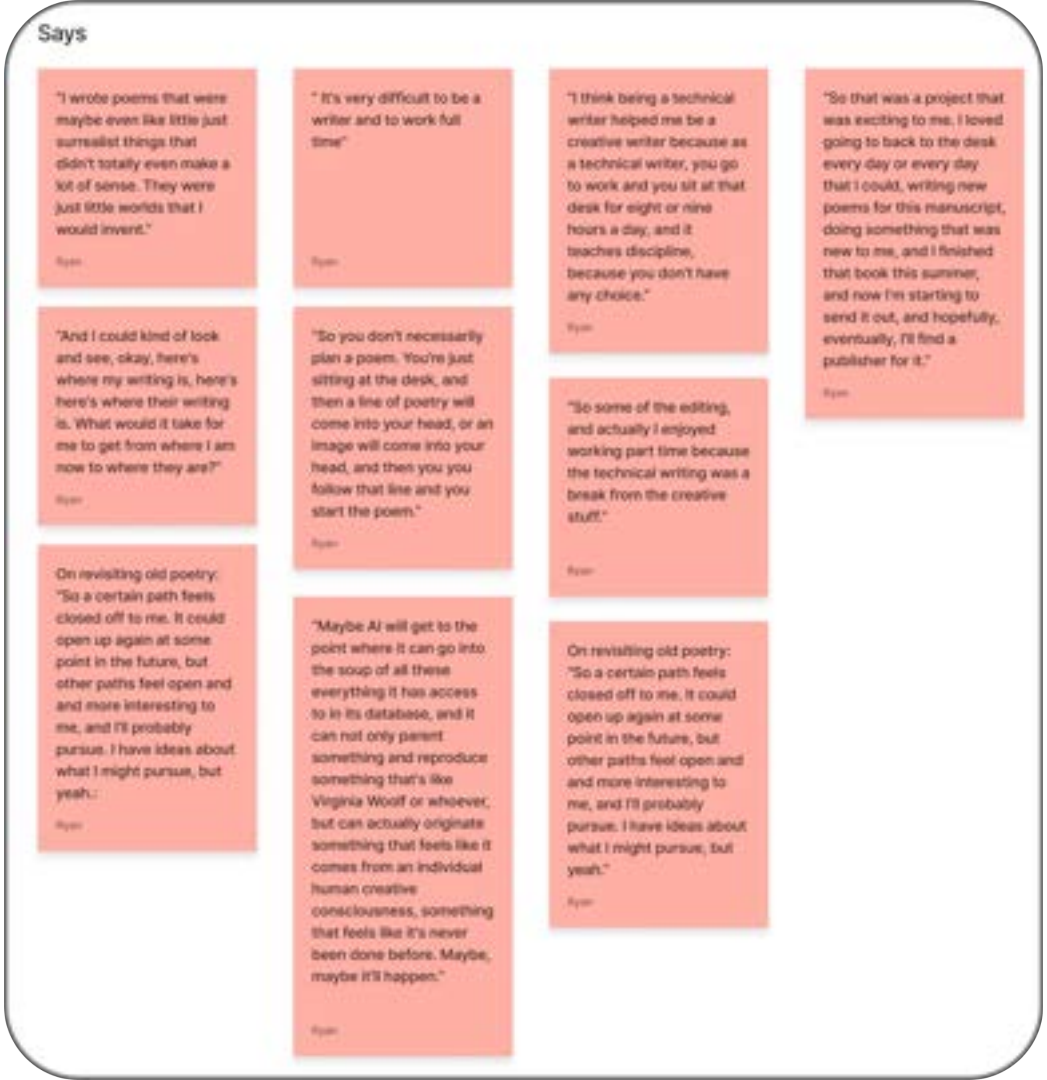
Feels sad, not angry, about what design work is, regarding the ownership she had in school and found rigid structure instead

Brian

Feels resentful that AI is forced on her by higher ups

Brian

Kevin
Empathy Map



Thinks

Thinks time to market does not matter

Ryan

Thinks it's hard to work full time and be creative

Ryan

Thinks young writers who use AI won't be able to struggle with the creative process

Ryan

Thinks writing is a habit

Ryan

Thinks that you can't sustain writing if you're not a novelist

Ryan

Thinks the best day at work is the days you know what you have to get done and aren't interrupted

Ryan

Thinks creative writing and technical writing require different skillsets

Ryan

Thinks or believes in Brenda Hillman "I don't experience writer's block, but I do experience moments where I feel like I'm writing far from the center of myself: those moments are frequent and painful"

Ryan

Thinks no one is at their best all the time

Ryan

Thinks meaningful work is when you feel like you've exceeded yourself in some way

Ryan

Does

- Has a file where he puts all of his poems that he can flip through — Ryan
- Talks a lot about his published works and the process by which they came to be — Ryan
- Goes on a lot of tangents about his life and how his experiences relate to his writing cycle — Ryan
- Sometimes gets distracted when writing when uninspired — Ryan
- Reads a lot of other work to find inspiration and also thinks back on his childhood to find inspiration — Ryan
- Teaches but also cuts down on his teaching hours when he plans on writing books — Ryan
- Translates epics from Costa Rican poet Eunice Odio — Ryan
- Uses AI to generate unimportant detail in his writings — Ryan
- Understands people who use AI in writing but believes that people need friction in writing — Ryan
- Challenges himself and finds new ways to surprise himself which he cannot achieve with revisiting failed poems — Ryan

Feels

Feels INTERESTED in the process of translation with AI

Ryan

Feels DISCIPLINED and will work every day for his job because of his background as a technical writer

Ryan

Feels that that because he has discipline, he has never felt burnout

Ryan

Feels that no matter what, his writing will always be a reflection of himself

Ryan

Feels that his brain is too fast for a physical notebook and that he can be more efficient on a laptop when writing poetry

Ryan

Feels that working full-time comes at the cost of having creative energy after your job

Ryan

Feels HOPEFUL that people have bad work because it's a part of the writing process

Ryan

Feels that the writing he produces immediately after finishing a novel is uninspired or secondary

Ryan

Feels HUMBLE in the presence of other writers, especially early on in his career

Ryan

Feels DISTRACTED when uninspired but overall has fluctuations with his inspiration depending on how recently he has finished writing a book

Ryan

Joseph
Empathy Map



Thinks

Thinks music is their biggest inspiration as opposed to other art

Ryan

Thinks meaningful work is all work that one puts effort into

Ryan

Thinks inspiration comes from instruction rather than community

Ryan

Thinks that comfort and fluency in a skill comes from consistent and daily repetition

Ryan

Thinks fluency in a program is translatable to other programs

Ryan

Thinks illustration should remain personal to retain value

Ryan

Thinks academics and creativity need a balance to both be pursued to their full potential

Ryan

Thinks creativity requires fresh experiences to flourish

Ryan

Thinks step-by-step tutorials are the most convenient and intuitive way to learn a skill

Ryan

Thinks difficulty/ challenge/struggle is necessary in the process of learning a skill

Ryan

Does

Joins clubs to maintain community in art

Ryan

Takes less courses to have more time to pursue art

Ryan

Watches YouTube videos to learn new skills

Ryan

Works on 2D and 3D animation in their free time

Ryan

Spends time drawing every day to exercise their muscle

Ryan

Enjoys learning new skills by navigating them with guidance

Ryan

Chases friction in the process of learning a new skill

Ryan

Took figure drawing classes to fix their anatomy and saw improvements right away

Brian

Listens to music for inspiration in art

Ryan

Practices both fine arts and cartoon arts

Ryan

Is chair of UC Berkeley's Anime Club

Ryan

Feels

Feels inspired by music rather than other art and visual stimuli; uses drawing as an output for their feelings

Ryan

Feels like they're constantly juggling time between academic and creative world

Ryan

Used to feel they had to sacrifice art for academics; now feels more at ease balancing the two

Ryan

Feels protective of their art: doesn't want AI finishing their pieces or training on their work

Ryan

Feels like they are capable of learning a new skill through guidance and repetition

Ryan

Feels that their creative side does not necessarily collaborate with their professional side

Ryan

Feels that they would not be satisfied if AI finished their work and made a beautiful piece because it had no struggle

Ryan

Feels that struggle is a key part of the learning process and that people have their own agency to learn

Ryan

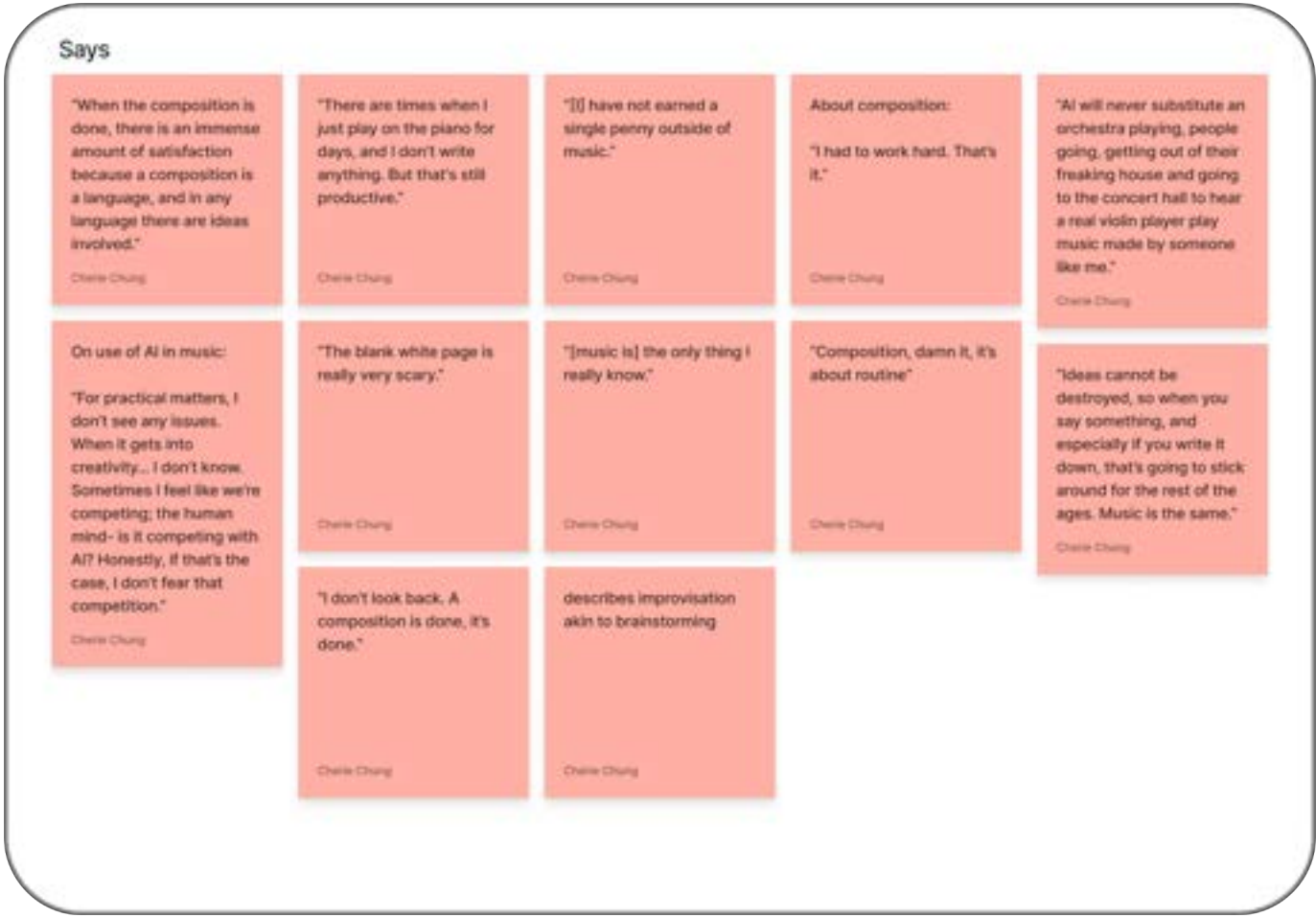
Feels that they can be burnt out in art if they engage in it too much/ treat it as work

Ryan

Feels that they value art and that it changes up their daily routine and allows them to appreciate certain things deeper

Ryan

Gary
Empathy Map



Thinks

Music composition isn't something that comes naturally

Charlie Chung

Inspiration [for composition] is something you find, not something that comes to you

Charlie Chung

believes that nothing he has done with music was a waste of time

Charlie Chung

There is a logic, and on that logic you apply your creativity

Charlie Chung

AI generated sounds/ music is just another tool or instrument-- similar to synthesized sounds

Charlie Chung

his ultimate goal is to make the better place

Charlie Chung

routine is the crucial key to success in music (instruments and composition alike)

Charlie Chung

believes that there is an existing logic in music

Charlie Chung

Creativity can also be learned

Charlie Chung

He thinks he is more creative now than he was as a teenager

Charlie Chung

Does

started learning composition to understand music better

Cherie Chung

often made sounds throughout the interview that resembled instruments

Cherie Chung

when stuck, he just sits in front of the piano playing

Cherie Chung

worked with the Grateful Dead on a music project for songs the band would play along with an orchestra; project started 15 years ago

Cherie Chung

will take 3 hours out of his day to sit at the piano and compose

Cherie Chung

he played notes on the piano when talking about how melodies never deteriorate unlike physical art

Cherie Chung

When stuck, he goes to his musical library, picks a score, looks it over, and tries to apply musical concepts

Cherie Chung

Spent a couple of years at CCRMA as a grad student before deciding electronic music wasn't for him

Cherie Chung

when composing, he turns off his phone and email notifications

Cherie Chung

a composition session does not *need* to involve writing; it can be just playing around with notes and melodies

Cherie Chung

Makes composition students bring a score to every lesson and asks what they learned from it

Brian

Feels

currently has mixed feelings about about AI

Cherie Chung

feels that AI music cannot compare to ones made by humans

Cherie Chung

feels that people still enjoy going in person to listen to music as a musical experience, more than listening to AI-generated music

Cherie Chung

He strongly feels that there is nothing about music that is not meaningful

Cherie Chung

still enjoys what he does--teaching students, going to rehearsals, composing music

Cherie Chung

feels like what he does is not work

Cherie Chung

hopes people, now and in the future, will want to go to see music played in person by real people composed by real composers

Cherie Chung

Feels intimidated by the blank page at the start of every new piece

Cherie Chung

Feels detached from his old pieces, reading them "like it's not mine anymore," sometimes impressed and sometimes embarrassed

Brian

Feels glad he tried CCRMA even though he didn't enjoy it

Brian

Feels no pull to retire, since music doesn't feel like work

Brian